

SoCG 2026 – 05/06/2026

Simplicial approximation to CW complexes with spherical Delaunay triangulations

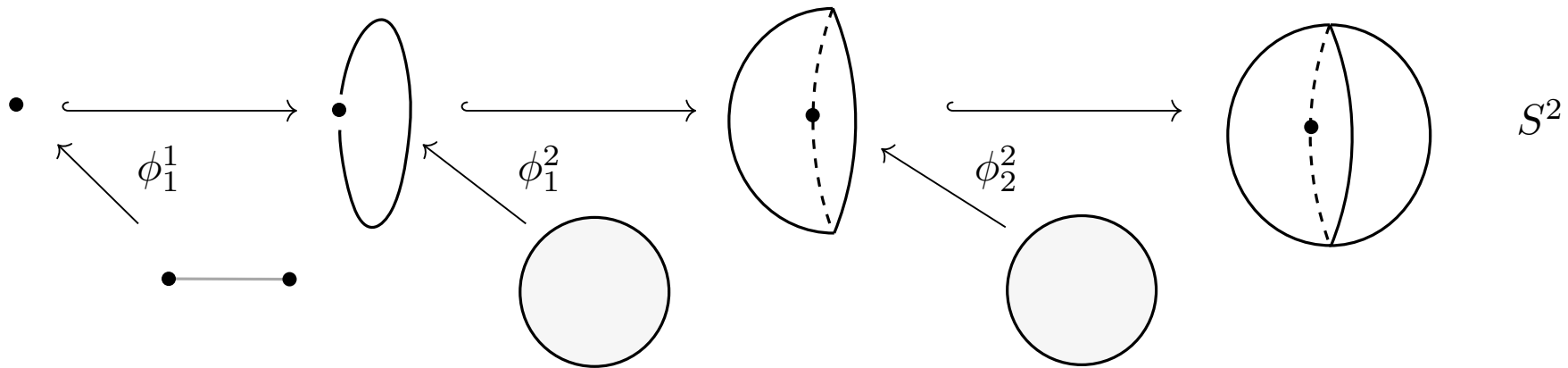
Raphaël Tinarrage – IST Austria



Space	Known triangulations	References
Real projective space $\mathbb{R}P^d$	$d \geq 1$	(Kühnel, 1987) (Adiprasito, Avvakumov, Karasev, 2022)
Complex projective space $\mathbb{C}P^d$	$d \geq 1$	(Sergeraert, 2010) (Sarkar, 2014) (Datta, Spreer, 2025)
Special orthogonal group $\mathrm{SO}(d)$	$d \leq 4$	$\mathrm{SO}(2) \simeq S^1, \quad \mathrm{SO}(3) \simeq \mathbb{R}P^3, \quad \mathrm{SO}(4) \simeq S^3 \times \mathrm{SO}(3)$
Grassmannian $\mathcal{G}(d, n)$	$d = 1$ or $n - 1$	$\mathcal{G}(1, n) \simeq \mathcal{G}(n - 1, n) \simeq \mathbb{R}P^{n-1}$ $\mathcal{G}(2, 4) \simeq (S^2 \times S^2)/\mathbb{Z}_2$
Oriented Grassmannian $\mathcal{G}^+(d, n)$	$d = 1$ or $n - 1$	$\mathcal{G}^+(1, n) \simeq \mathcal{G}^+(n - 1, n) \simeq S^{n-1}$ $\mathcal{G}^+(2, 4) \simeq S^2 \times S^2$
Stiefel manifold $\mathcal{V}(d, n)$	$d = 1$ or $n \leq 4$	$\mathcal{V}(1, n) \simeq S^{n-1}, \quad \mathcal{V}(n - 1, n) \simeq \mathrm{SO}(n), \quad \mathcal{V}(n, n) \simeq \mathrm{O}(n)$ $\mathcal{V}(2, 4) \simeq S^3 \times S^2, \quad \mathcal{V}(2, 8) \simeq S^7 \times S^6$

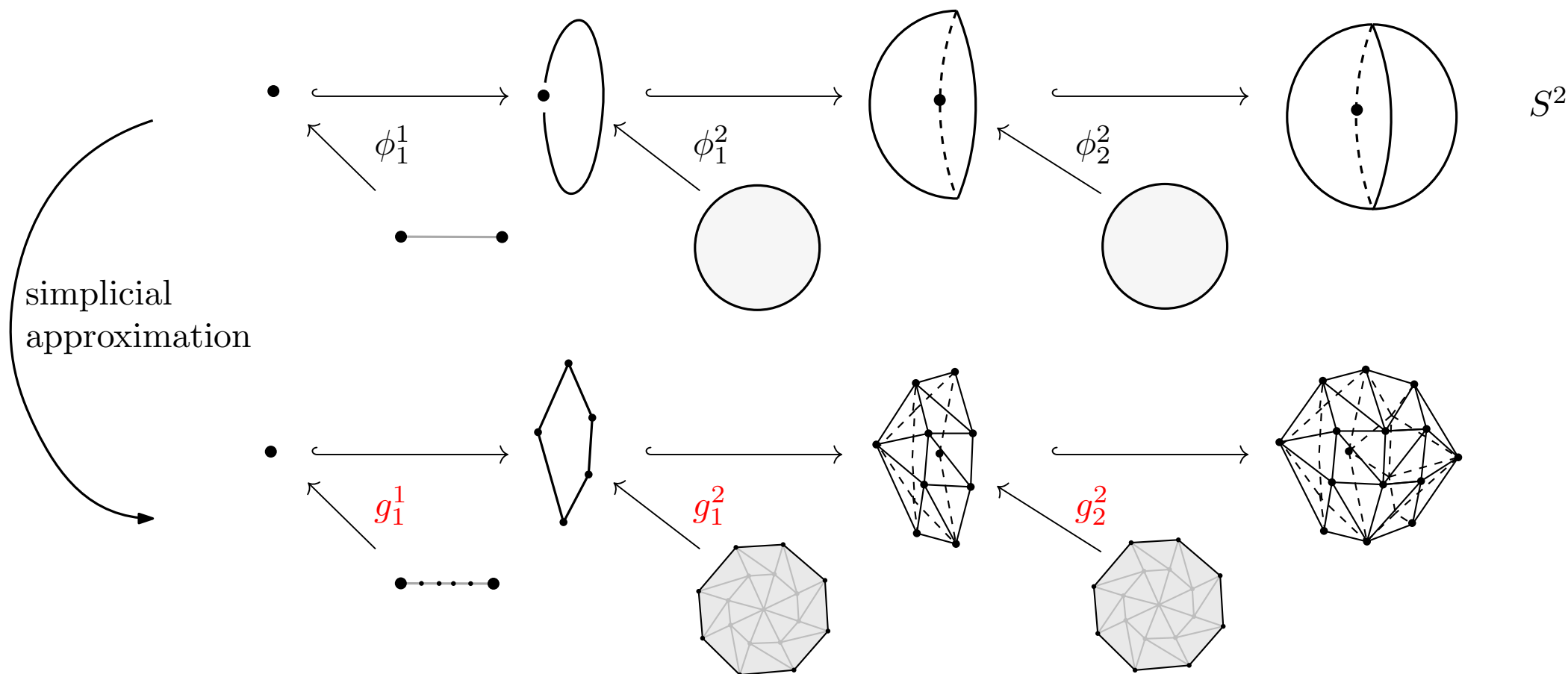
CW structure on X : Sequence of gluings of balls along their boundaries, via gluing maps $\phi_i^d: S^{d-1} \rightarrow X$.

$$X = \underbrace{\{\text{pt}\} \cup \{\text{pt}\}}_{\text{points}} \cup \dots \cup \underbrace{\phi_1^1 B^1 \cup \phi_2^1 B^1}_{\text{1-balls}} \cup \dots \cup \underbrace{\phi_1^2 B^2 \cup \phi_2^2 B^2}_{\text{2-balls}} \cup \dots$$

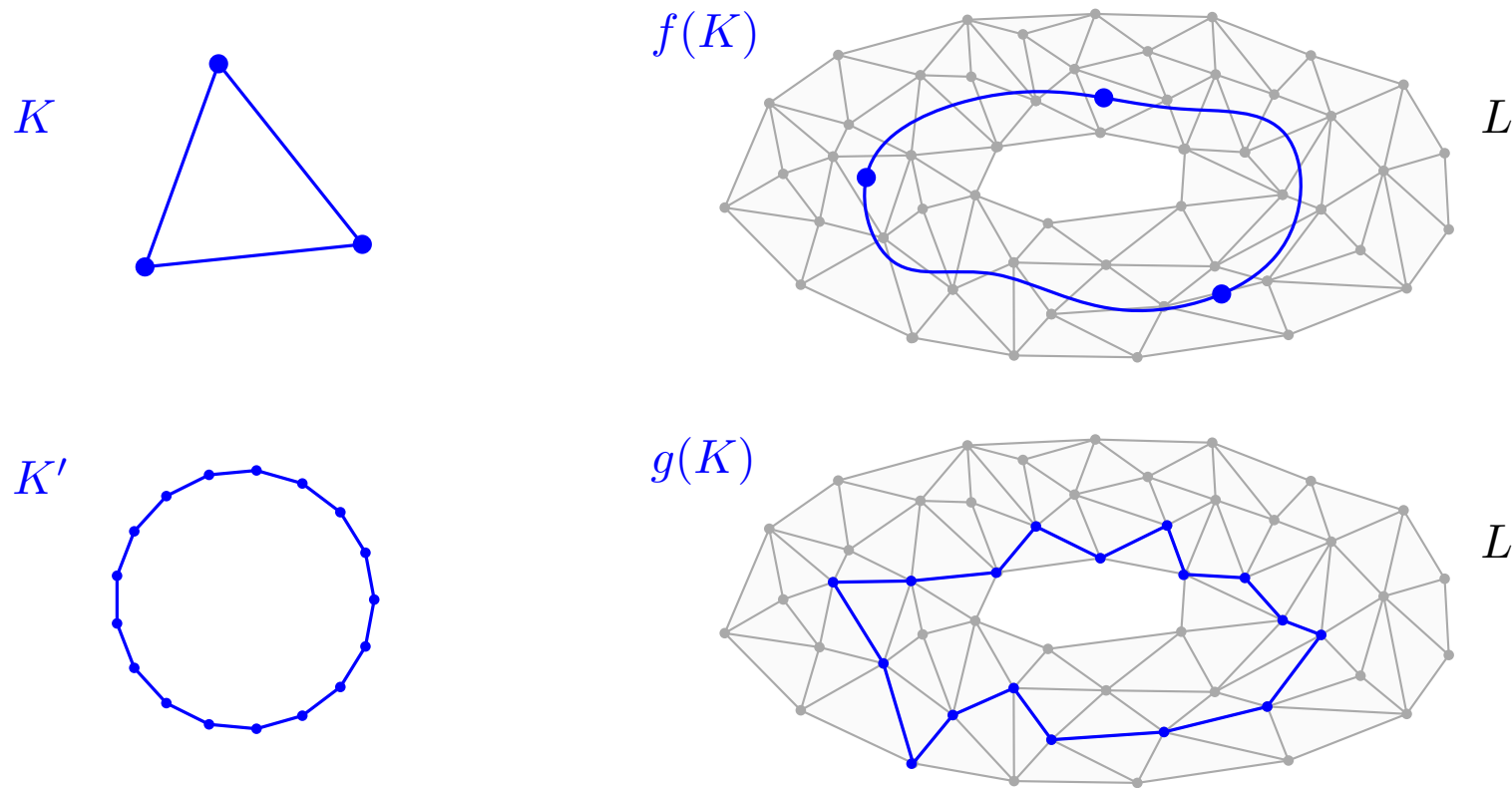


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Simplicial Approximation Theorem: Consider simplicial complexes K, L and a continuous map $f: |K| \rightarrow |L|$. After sufficiently many barycentric subdivisions of K , there exists a simplicial map $g: K \rightarrow L$ homotopic to f .

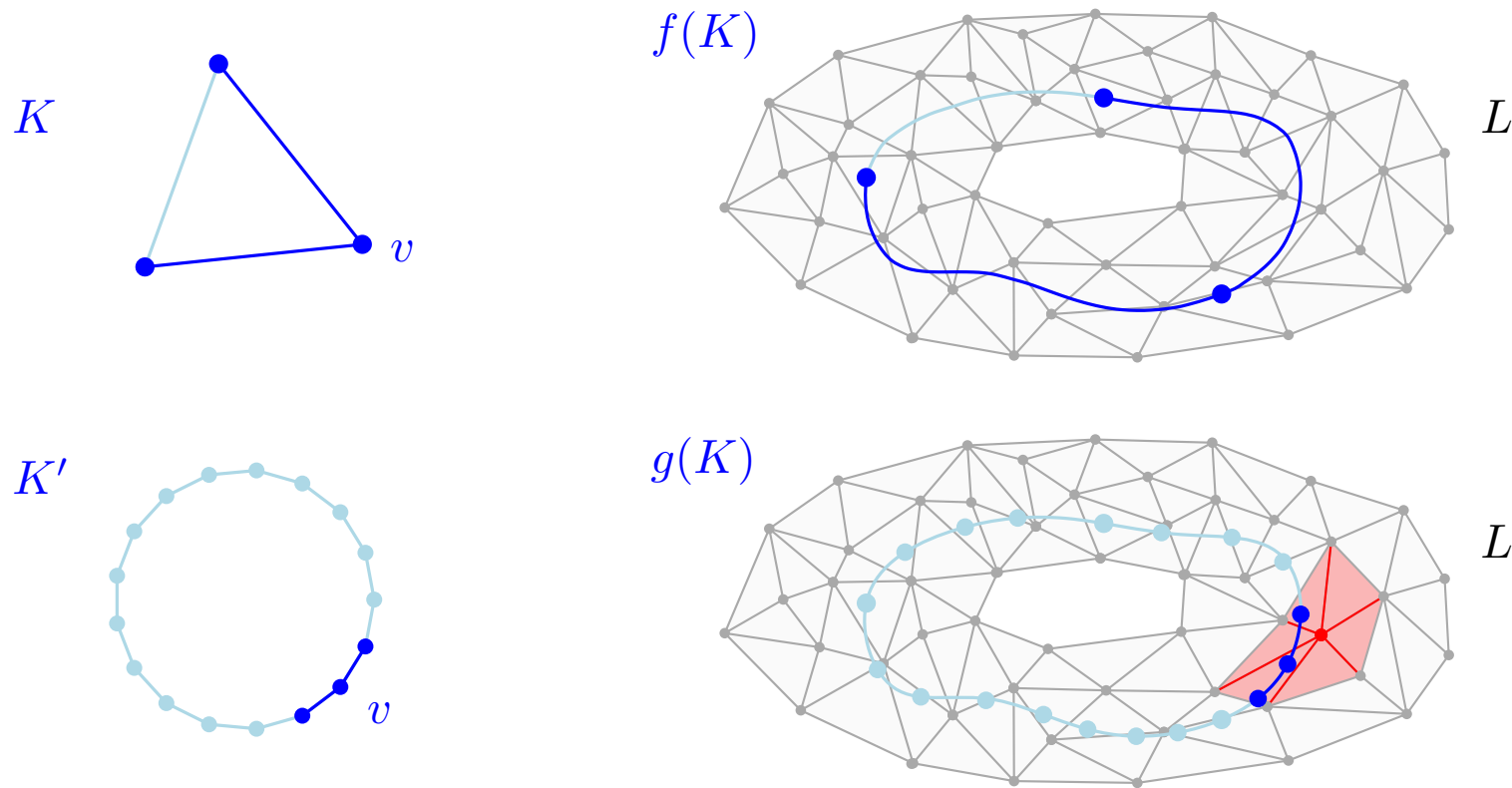


Proof: Say that f satisfies the **star condition** if

$$\forall v \in V(K), \exists w \in V(L) \text{ s.t. } f(|\overline{\text{St}}(v)|) \subseteq |\text{St}(w)|.$$

If this holds (and it will), we can build a simplicial map g by selecting for each v a corresponding w .

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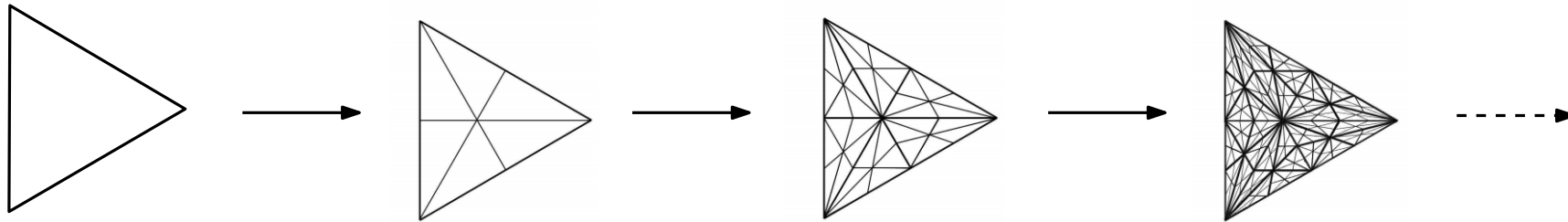


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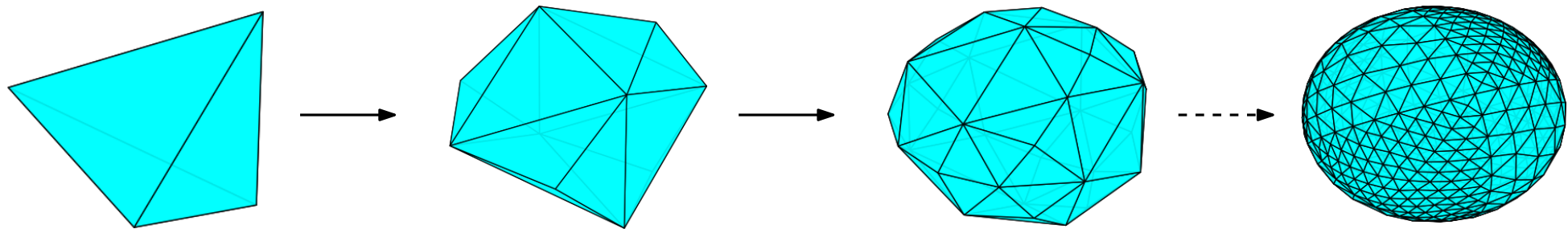
Barycentric subdivision:

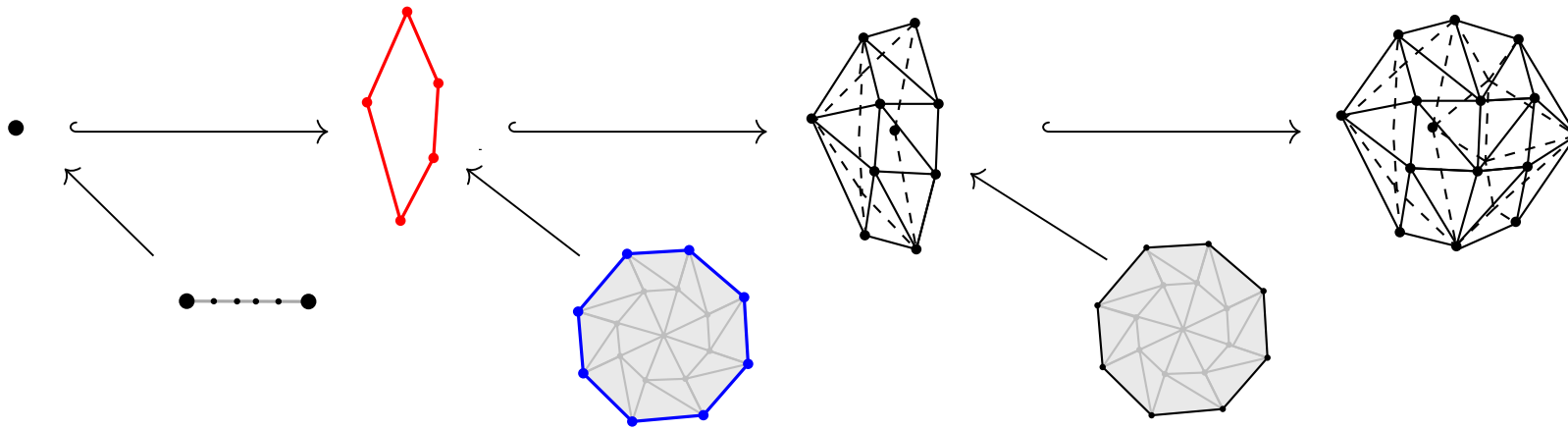


Problem: The barycentric subdivision of Δ^d is a complex with $2^{d+1} - 1$ vertices and $(d + 1)!$ simplices.

Delaunay refinement:

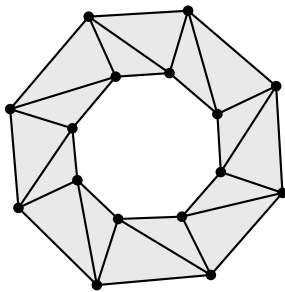
Build spherical Delaunay complexes $\text{Del}(X_0)$, $\text{Del}(X_1)$, $\dots$ by introducing **Steiner points** at each step.



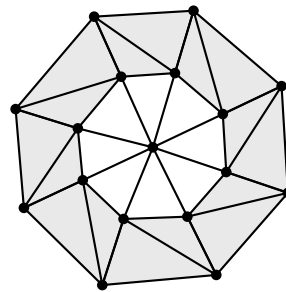


Say we found a simplicial map $g: K \rightarrow L$.

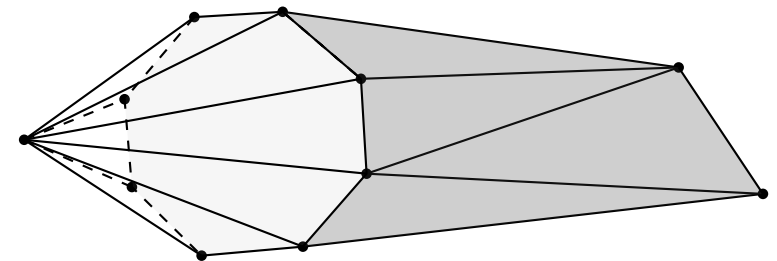
Next step: build the **mapping cone** of g .



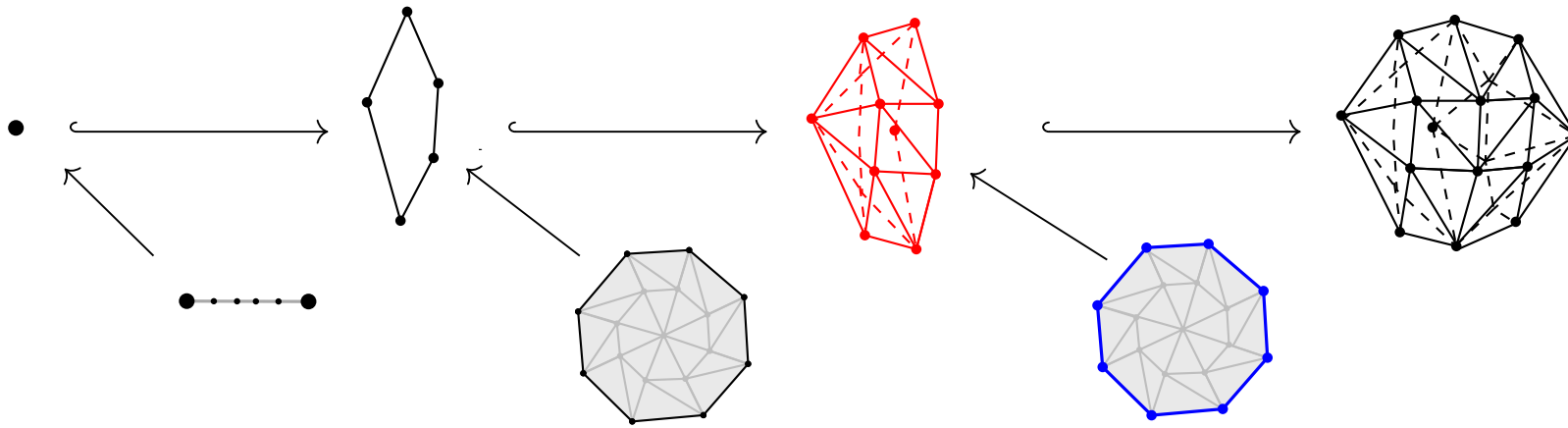
Triangulate $|K| \times [0, 1]$



Cone the inner layer

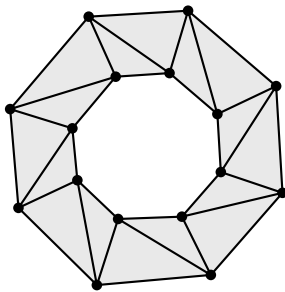


Glue the outer layer onto L

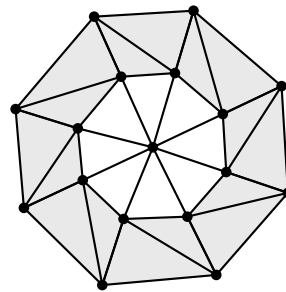


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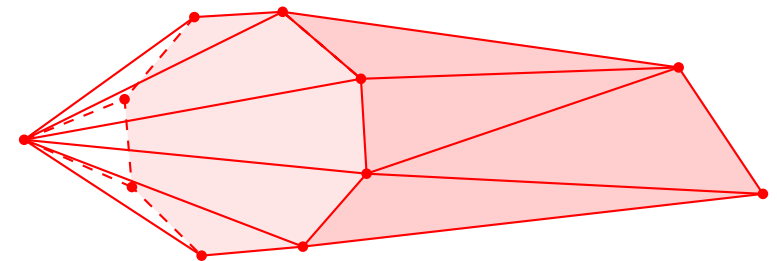
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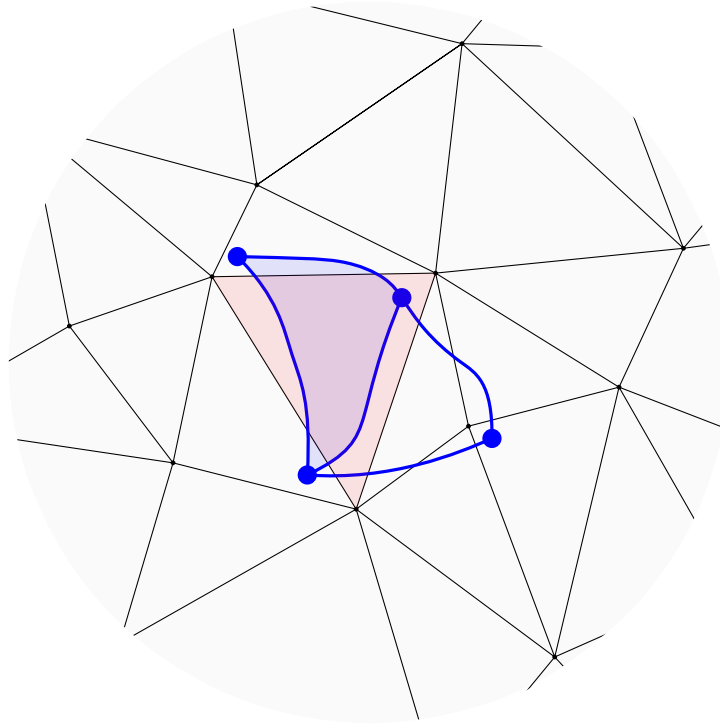


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Glue the outer layer onto L

The map $f: |K| \rightarrow |L|$ satisfies the **star condition** if $\forall v \in V(K), \exists w \in V(L)$ s.t. $f(|\overline{\text{St}}(v)|) \subseteq |\text{St}(w)|$.

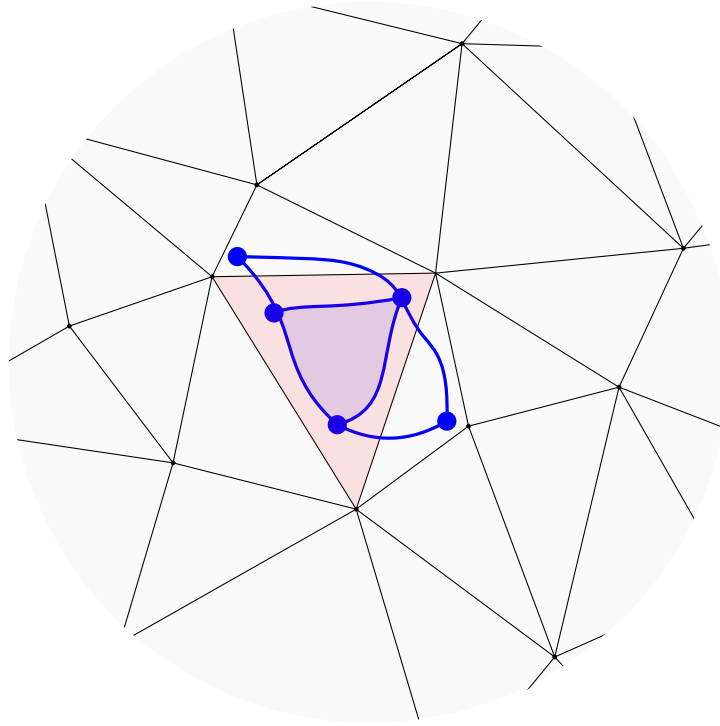


Let g be a simplicial approximation.

If g maps a facet $\sigma \in K$ to a facet $\tau \in L$, then $f(|\sigma|) \subset |\tau|$.

Problem: Each facet of L requires $\dim(L) + 1$ vertices in K .

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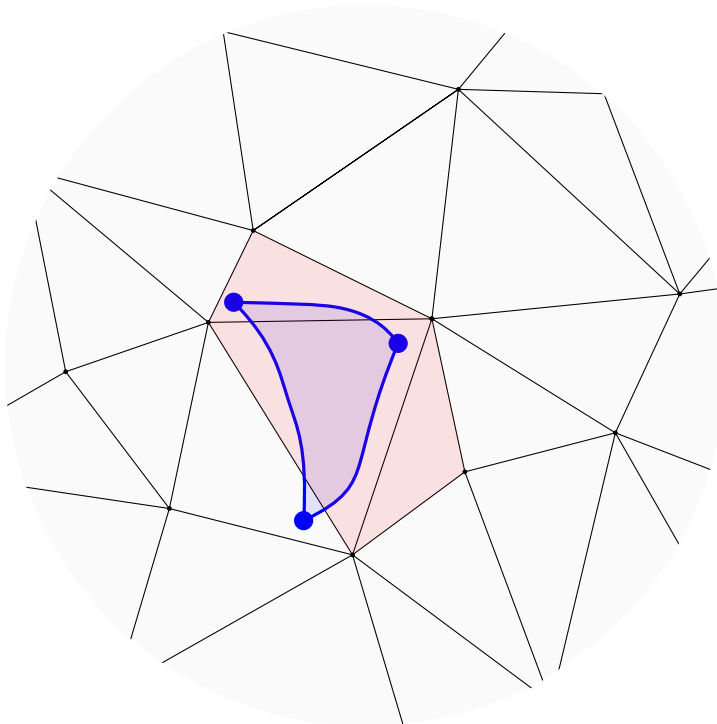
Problem: Each facet of L requires $\dim(L) + 1$ vertices in K .

Split simplicial approximation into two problems:

Geometric feasibility

For each facet $\sigma \in K$, find admissible facets in L .

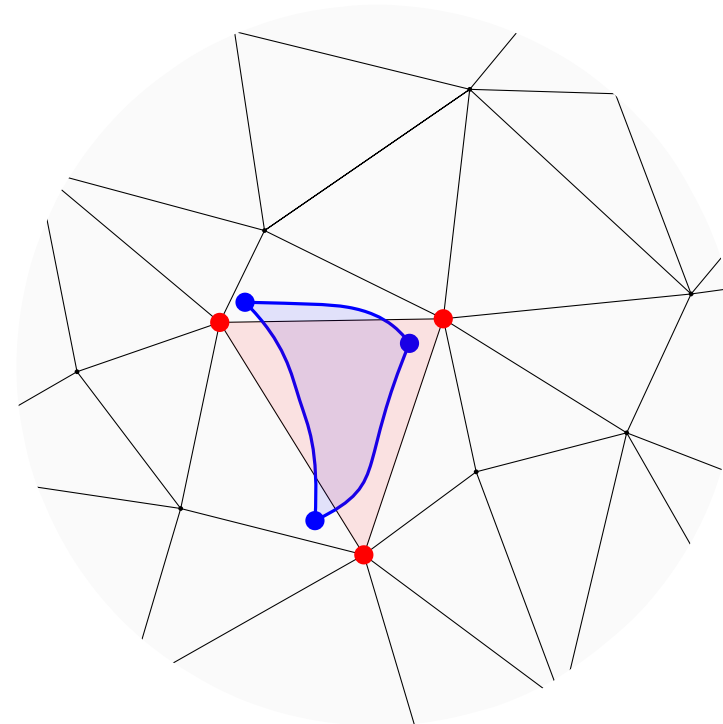
→ Define a homotopy via equiconnecting maps.



Combinatorial feasibility

Given admissible facets, find a simplicial map.

→ Solve the list homomorphism problem.



Idea (Dugundji, 1965): Draw paths $t \mapsto \Pi(x, y, t)$ in the space.

A space Y is **locally equiconnected** if there exists a neighborhood $U \subset Y \times Y$ of the diagonal and a continuous map $\Pi: U \times [0, 1] \rightarrow Y$ such that for all $(x, y) \in U$ and $t \in [0, 1]$,

$$\Pi(x, y, 0) = x, \quad \Pi(x, y, 1) = y, \quad \Pi(x, x, t) = x.$$

Observation: Two maps $f, g: X \rightarrow Y$ are homotopic whenever they are U -close, i.e.,

$$(f(x), g(x)) \in U \text{ for all } x \in X.$$

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Proposition (Equiconnecting maps on simplicial mapping cones):

Let $|L|$ be endowed with an equiconnecting map (Π_L, U_L) . Then $|L \cup_g B(K)|$ admits an equiconnecting map $\Pi: U \times [0, 1] \rightarrow |L \cup_g B(K)|$ such that $(x, y) \in U$ in each of these cases:

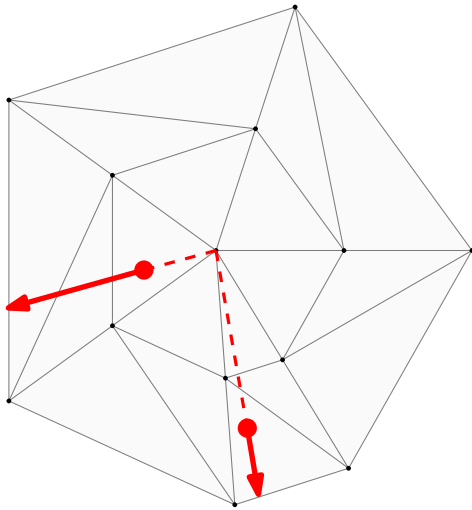
Boundary: $x, y \in |L|$ and $(x, y) \in U_L$.

Mixed: $x \in |L|$, $y \in |B(K)|$, $\|y\| \neq 0$ and $(x, y/\|y\|) \in U_L$.

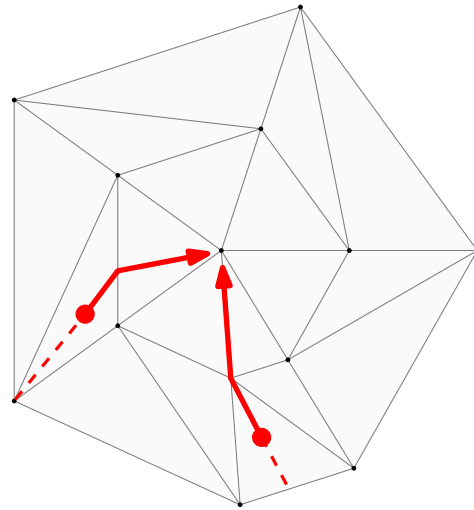
Interior: for $x, y \in |B(K)|$,

1. $\|x\| = 0$, or
2. $\|x\|, \|y\| \leq 1/2$, or
3. $x/\|x\| \neq -y/\|y\|$ and $(x/\|x\|, y/\|y\|) \in U_L$.

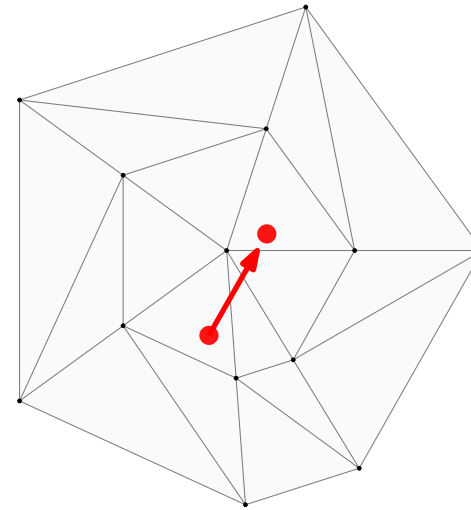
Admissible paths (that descend to the quotient).



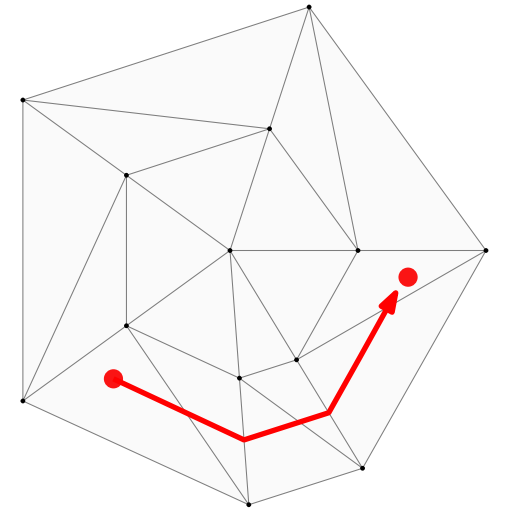
Ray away from the origin



Climb toward the origin

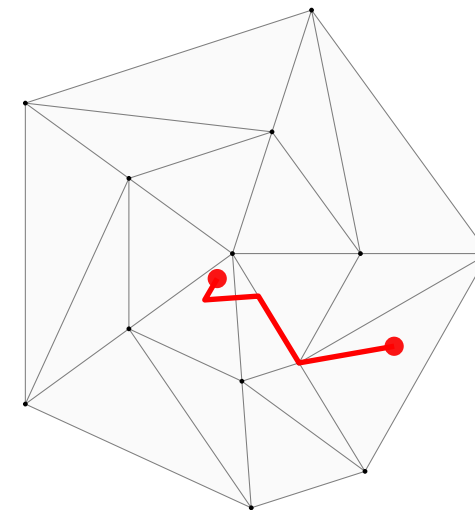
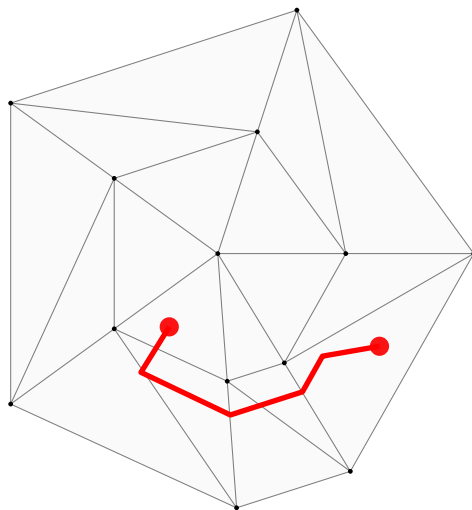


Straight path in inner layer



Circular arc in outer layer
(points of equal norm)

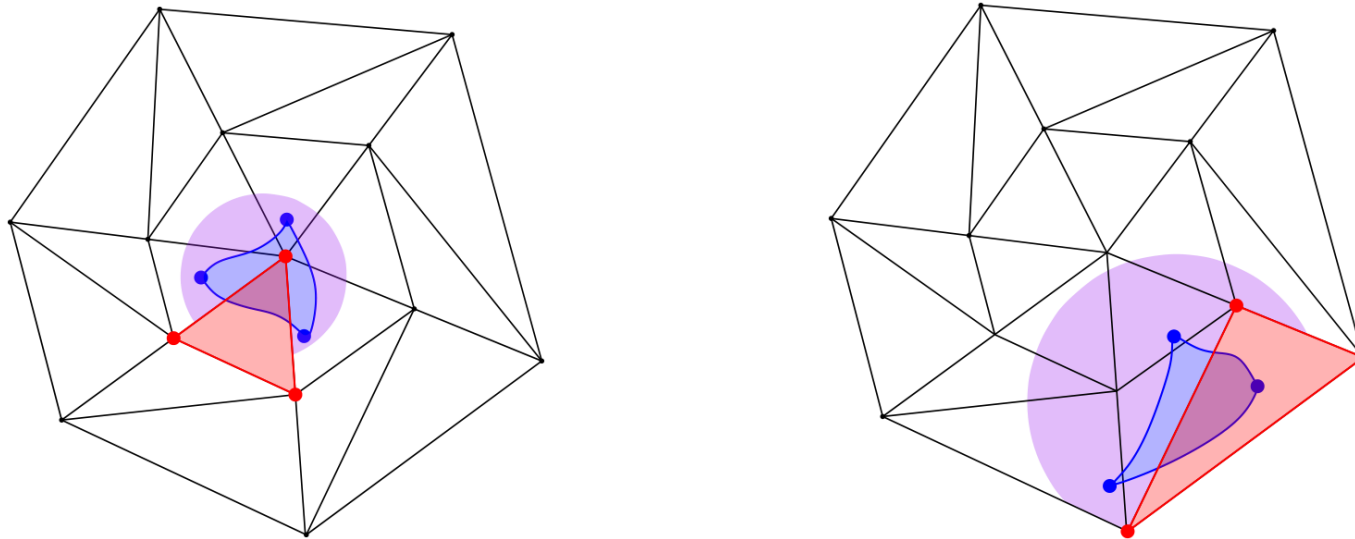
They are combined to define paths on the ball.



Let $f: |K| \rightarrow |L|$, where $|L|$ is endowed with an equiconnecting map (Π, U) .

We seek a U -close simplicial map $g: K \rightarrow L$, i.e., $(f(x), |g|(x)) \in U$ for all $x \in |K|$.

For each facet $\sigma \in K$, we check whether $f(|\sigma|)$ is U -close to a simplex of L .



Lemma: Let $\sigma = [v_0, \dots, v_d]$ be a simplex in $\mathbb{R}^n$ and $f: |\sigma| \rightarrow \mathbb{R}^m$ a λ -Lipschitz map. For $I \subset \{0, \dots, d\}$, let

$$c_I = \frac{1}{|I|} \sum_{i \in I} f(v_i) \quad \text{and} \quad r_I = \max_{0 \leq k \leq d} \frac{1}{|I|} \sum_{i \in I} \|v_k - v_i\|.$$

Then $f(|\sigma|)$ is included in the closed ball $B(c_I, \lambda r_I)$.

We obtain, for each facet $\sigma \in K$, a collection of admissible simplices $\tau_1, \tau_2, \dots$ of L .

From these, we derive for each vertex $v \in V(K)$ a set $\ell(v) \subset V(L)$ of admissible images, thus defining a map

$$\ell: V(K) \rightarrow \mathcal{P}(V(L))$$

LHom

Input: Simplicial complexes K, L and a subset $\ell(v) \subset V(L)$ for each $v \in V(K)$.

Output: A simplicial map $g: K \rightarrow L$ with $g(v) \in \ell(v)$ for all $v \in V(K)$.

Polynomial-time solvable if L is a bi-arc graph, NP-complete otherwise (Feder, Hell, Huang, 2003).

Software: MiniSAT, OR-Tools, Z3, ...

If the problem is not feasible, we subdivide L .

Conclusion

- Implementation of a certified algorithm for triangulating arbitrary CW complexes.
- Instead of the star condition, it uses the planner condition + **LHom** solver.
- Cells of dimension up to 5 are easily glued.
- In particular, we triangulated $\mathcal{G}(2,4)$ and $\mathcal{V}(2,4)$.

